

You must show your work to receive. Each problem is worth 4 pts unless stated otherwise.

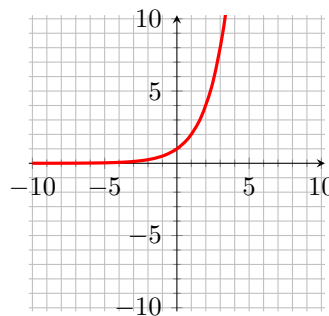
1. Use the function, $f(x) = 2^x$, to complete parts 1a–1d.

a. Graph $f(x)$. Label the y -int and at least one other point.

b. Find the domain of $f(x)$.
 $(-\infty, \infty)$

c. Find the range of $f(x)$.
 $(0, \infty)$

d. Find the horizontal asymptote of $f(x)$.
 $y = 0$



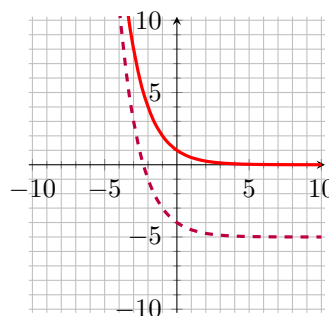
2. Use the function, $f(x) = \left(\frac{1}{2}\right)^x$, to complete parts 2a–2b.

a. Graph $f(x)$. Label the y -int and at least one other point.

b. Find the horizontal asymptote of

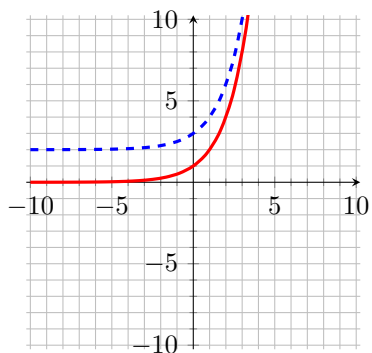
$$g(x) = \left(\frac{1}{2}\right)^x - 5.$$

$$y = -5 \text{ (} g(x) \text{ is the dashed curve)}$$

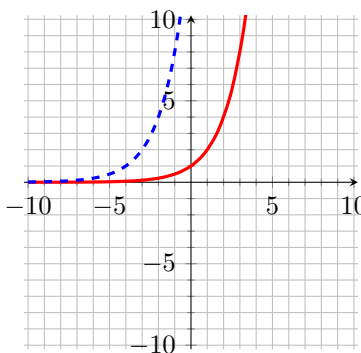


3. Use the graph of b^x in parts 3a–3c to graph the given transformation.

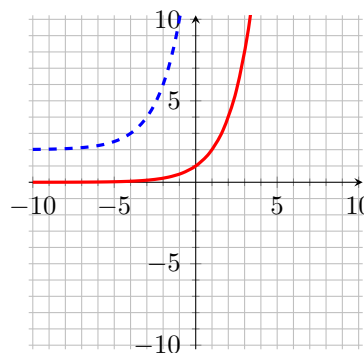
a. Graph $b^x + 2$



b. Graph b^{x+4}



c. Graph $b^{x+4} + 2$



Use the formulas, $A = P(1+r)^t$, $A = P(1 + \frac{r}{n})^{nt}$, or $A = Pe^{rt}$ to problems 4–6.

4. [6 points] Find the accumulated value of an investment of \$850 at 9% interest compounded quarterly after 10 years.

$$P = 850, r = .09, t = 10, \text{ and } n = 4. \text{ So } A = \left(1 + \frac{.09}{4}\right)^{4 \cdot 10} \approx \$2069.911$$

5. [6 points] Bob buys a house for \$150000. The value of the house grows at a rate of 4.5% compounded monthly. Find how much the house is worth after 15 years.

$$P = 150000, r = .045, t = 15, \text{ and } n = 12. \text{ So } A = \left(1 + \frac{.045}{12}\right)^{12 \cdot 15} \approx \$272504.505$$

6. [6 points] Suppose Jane has \$2000 to invest. Which investment yields the greater return over 10 years: 7% compounded monthly or 6.5% compounded continuously? Justify your answer.

- 7% compounded monthly

$$P = 2000, r = .07, t = 10, \text{ and } n = 12. \text{ So } A = \left(1 + \frac{.07}{12}\right)^{12 \cdot 10} \approx \$4019.323$$

- 6.5% compounded continuously

$$P = 2000, r = .065, t = 10. \text{ So } A = 2000e^{.065 \cdot 10} \approx \$3831.082$$

So 7% compounded monthly is better.

7. Write the expression in its equivalent *exponential* form for parts 7a–7c.

a. $\log_3 x = 9$
 $3^9 = x$

b. $\ln x = 5$
 $e^5 = x$

c. $\log_x 2 = c$
 $x^c = 2$

8. Write the expression in its equivalent *logarithmic* form for parts 8a–8c.

a. $5^x = 125$

b. $x^{12} = 7$

c. $x^{-2} = \frac{1}{16}$
 $\log_x \frac{1}{16} = -2$

$$\log_5 125 = x$$

$$\log_x 7 = 12$$

9. [3 points each] Evaluate or simplify the expression without using a calculator for parts 9a–9f.

a. $\log_5 5 = 1$

c. $\log_2 2^{10} = 10$

e. $\log_5 25 = 2$ since $5^2 = 25$

b. $e^{\ln 2} = 2$

d. $4^{\log_4 11} = 11$

f. $\log 10 = 1$ since $10^1 = 10$

10. Find the domain of the logarithmic function for parts 10a–10b.

a. $\log_{13} x$

$$\text{Domain: } (0, \infty)$$

b. $\ln(x - 2)$

$$x - 2 > 0 \Rightarrow x > 2$$

$$\text{Domain: } (2, \infty)$$